# Formula 110 Lab Notes

August 28 - September 1, 2026

## 8/28/26 @ 12pm

**Participants and contributions:** Lucas, Zhi

**Question or objective**

Create plan for researching approaches

**What we investigated or changed**

Decided that Zhi will investigate an Evolutionary Computation and Neuroevolution approach, Lucas will investigate Hybrid approach of pre-training model based on the seed tracks.

**Evidence**

- AI-agent assistance: We both began making plans with Codex to identify the validity of our ideas.

**What we observed**

N/A

**Next steps**

We will reconvene on Monday to discuss our progress and compare results from our approaches

## 8/29/26

**Participants and contributions:** Zhi

**Question or objective**

Exploring creating a Neuroevolution algorithm to improve lap time

**What we investigated or changed**

I created a minimal viable model that completes first lap in 28 seconds, it takes the sensors and try to always stay in the middle of the track to safely complete the lap without any damage. I then use that model as the parent/seed for the first generation to save training time. I first explore using cross-entropy method for version 1 and 2.

**Evidence**

- AI-agent assistance: I had codex generate a simple `minimum_viable` mode and ran it. Then create a initial CEM model to try to get faster lap times.

**What we observed**

improvements were immediate, 28 down to 10 seconds in just 2 versions (around 140 generations).

**Next steps**

However, after seeing those race car machine learning videos, I want to attempt soemthing like that. I'll take the current result and continue with a genetic algorithm + mutations approach. Also another concern that sparked this change, is while time has gotten better, after manually reviewing the line the racecar took, it just optimized the center line used by minimal viable racecar, which I feel like is going to converge really quickly.

## 8/29/26 @ 12pm

**Participants and contributions:** Lucas

**Question or objective**

Developing training mechanism for the pre-training approach

**What we investigated or changed**

Created a privileged offline cross-entropy method search training mechanism. Starts with a broad, random distribution of throttle/steer action sequences, runs simulations, keeps the best-performing candidates, repeats based on the previous best-performing candidates.

**Evidence**

- AI-agent assistance: Developed the training code. I worked with the agent to assess the training mechanism and identify potential improvements.
- Commits or code: `scripts/hybrid_training_v1/`
- Experiment output: Initial trials produced a car that did not complete a full lap around the track but sufficient initial training completed ~1 full lap in 30s locally.

**What we observed**

See experiment output above

**Decision and rationale / next steps**

Start a longer-term training session, as more specific optimization is needed

## 8/30/26

**Participants and contributions:** Zhi

**Question or objective**

Improve lap time by switching from CEM to a genetic algorithm with mutations.

**What we investigated or changed**

CEM was narrowing toward one similar, conservative set of parameters, so I switched to a GA that keeps the best controllers, combines their parameters, and adds random mutations. This lets the search maintain more variety and try different racing lines.

**Evidence**

- AI-agent assistance: I used Codex to help implement and test the GA search.
- The GA improved the best lap from about 9.55 seconds to 8.98 seconds while keeping the controller safe across the training seeds.

**What we observed**

The GA found a faster racing line than CEM because it continued exploring different parameter combinations instead of quickly settling around one average solution.

**Next steps**

Continue tuning with GA and test the best controller on more seeds. I have a couple of ideas how to improve time if the model converges to a time with no improve after multiple generations.

## 8/30/26 @ 8pm

**Participants and contributions:** Lucas

**Question or objective**

Is pre-training the best approach?

**What we investigated or changed**

Separated the map into "sectors" to determine the car's best behavior at each point on the track

**Evidence**

- AI-agent assistance: AI agent helped me analyze the result from the portion of the training that did complete, I determined that it wasn't produced a competitive-enough result.
- Commits or code: `scripts/hybrid_training_v1/`
- Experiment output: Started the traning, it ran out of memory and crashed my laptop after several hours (presumably ran out of memory)

**What we observed**

See experiment output above

**Decision and rationale / next steps**

Pre-training the model based on every possible track position is likely not the most efficient approach and would completely break if exposed to a new track. Next steps are to abandon this approach and try using a model that is less deterministic.

## 8/31/26

**Participants and contributions:** Zhi

**Question or objective**

Try mutating different controller parameters to continue improving the GA's lap time.

**What we investigated or changed**

With help from Codex, I adjusted which parameters the GA could mutate and changed their allowed ranges across several versions.

**Evidence**

- AI-agent assistance: Codex helped choose and adjust the parameter groups and mutation ranges.
- The best lap times eventually stopped improving and converged at about 7.6 seconds.

**What we observed**

Changing the mutation parameters produced smaller improvements over time, but the GA eventually reached a plateau that it could not break.

**Next steps**

I will probably add some imitation learning by manually driving and recording human-controlled laps, then use those sensor and control examples to help the controller learn a different racing strategy.

## 9/1/26

**Participants and contributions:** Zhi

**Question or objective**

Test whether drifting learned from manually controlled laps could improve the 7.6-second best lap.

**What we investigated or changed**

I introduced drifting behavior based on the human-controlled runs and allowed the controller to take a small amount of damage in exchange for more speed.

**Evidence**

- The best lap time improved from about 7.6 seconds to 7.2 seconds.

**What we observed**

The new controller is faster, but I am not yet sure whether its extra damage makes it a better replacement for the safer 7.6-second controller.

**Next steps**

Meet with Lucas and try to combine what we have developed.